Research Notes on Image to image translation

Image to image translation

>> Section 1

Level-set methods The level-set method was initially proposed to track moving interfaces by Dervieux and Thomasset in 1979 and 1981 and was later reinvented by Osher and Sethian in 1988. This has spread across various imaging domains in the late 1990s. It can be used to efficiently address the problem of curve/surface/etc. propagation in an implicit manner. The central idea is to represent the evolving contour using a signed function whose zero corresponds to the actual contour. Then, according to the motion equation of the contour, one can easily derive a similar flow for the implicit surface that when applied to the zero level will reflect the propagation of the contour. The level-set method affords numerous advantages: it is implicit, is parameter-free, provides a direct way to estimate the geometric properties of the evolving structure, allows for change of topology, and is intrinsic. It can be used to define an optimization framework, as proposed by Zhao, Merriman and Osher in 1996. One can conclude that it is a very convenient framework for addressing numerous applications of computer vision and medical image analysis. Research into various level-set data structures has led to very efficient implementations of this method.

>> Section 2

α (1 – δ (■ i – ■ initial i) + β Σ q ∊ N (i) (1 – δ (■ i , ■ q (i))) . {\displaystyle \alpha (1-\delta (\ell _{i}-\ell _{{\text{initial }}i})+\beta \Sigma _{q\in N(i)}(1-\delta (\ell _{i},\ell _{q(i)})).}

>> Section 3

argmin u , K ■ γ | K | + μ ∫ K C | ∇ u | 2 d x + ∫ (u – f) 2 d x . {\displaystyle \operatorname {argmin} _{u,K}\gamma |K|+\mu \int _{K^{C}}|\nabla u|^{2}\,dx+\int (u-f)^{2}\,dx.}

>> Section 4

Markov random fields The application of Markov random fields (MRF) for images was suggested in early 1984 by Geman and Geman. Their strong mathematical foundation and ability to provide a global optimum even when defined on local features proved to be the foundation for novel research in the domain of image analysis, de-noising and segmentation. MRFs are completely characterized by their prior probability distributions, marginal probability distributions, cliques, smoothing constraint as well as criterion for updating values. The criterion for image segmentation using MRFs is restated as finding the labelling scheme which has maximum probability for a given set of features. The broad categories of image segmentation using MRFs are supervised and unsupervised segmentation.

>> Section 5

Edge detection Edge detection is a well-developed field on its own within image processing. Region boundaries and edges are closely related, since there is often a sharp adjustment in intensity at the region boundaries. Edge detection techniques have therefore been used as the base of another segmentation technique. The edges identified by edge detection are often disconnected. To segment an object from an image however, one needs closed region boundaries. The desired edges are the boundaries between such objects or spatial-taxons. Spatial-taxons are information granules, consisting of a crisp pixel region, stationed at abstraction levels within a hierarchical nested scene architecture. They are similar to the Gestalt psychological designation of figure-ground, but are extended to include foreground, object groups, objects and salient object parts. Edge detection methods can be applied to the spatial-taxon region, in the same manner they would be applied to a silhouette. This method is particularly useful when the disconnected edge is part of an illusory contour Segmentation methods can also be applied to edges obtained from edge detectors. Lindeberg and Li developed an integrated method that segments edges into straight and curved edge segments for parts-based object recognition, based on a minimum description length (MDL) criterion that was optimized by a split-and-merge-like method with candidate breakpoints obtained from complementary junction cues to obtain more likely points at which to consider partitions into different segments.

>> Section 6

Compression-based methods Compression based methods postulate that the optimal segmentation is the one that minimizes, over all possible segmentations, the coding length of the data. The connection between these two concepts is that segmentation tries to find patterns in an image and any regularity in the image can be used to compress it. The method describes each segment by its texture and boundary shape. Each of these components is modeled by a probability distribution function and its coding length is computed as follows:

>> Section 7

∇ 2 f (x , y) = f (x + 1 , y) + f (x – 1 , y) + f (x , y + 1) + f (x , y – 1) – 4 f (x , y) {\displaystyle \nabla ^{2}f(x,y)=f(x+1,y)+f(x-1,y)+f(x,y+1)+f(x,y-1)-4f(x,y)}

>> Section 8

Here λ ∊ Λ {\displaystyle \lambda \in \Lambda } , the set of all possible labels. 3. M step: The established relevance of a given feature set to a labeling scheme is now used to compute the a priori estimate of a given label in the second part of the algorithm. Since the actual number of total labels is unknown (from a training data set), a hidden estimate of the number of labels given by the user is utilized in computations.

>> Section 9

Pick K cluster centers, either randomly or based on some heuristic method, for example K-means++ Assign each pixel in the image to the cluster that minimizes the distance between the pixel and the cluster center Re-compute the cluster centers by averaging all of the pixels in the cluster Repeat steps 2 and 3 until convergence is attained (i.e. no pixels change clusters) In this case, distance is the squared or absolute difference between a pixel and a cluster center. The difference is typically based on pixel color, intensity, texture, and location, or a weighted combination of these factors. K can be selected manually, randomly, or by a heuristic. This algorithm is guaranteed to converge, but it may not return the optimal solution. The quality of the solution depends on the initial set of clusters and the value of K. The Mean Shift algorithm is a technique that is used to partition an image into an unknown apriori number of clusters. This has the advantage of not having to start with an initial guess of such parameter which makes it a better general solution for more diverse cases.